MAIS PAC

IPhO 02.09: Oscillations and Resonance · Worked Solutions

Oscillations and Resonance: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IPhO questions or marking schemes. Parameters are supplied locally; an exam’s constants take precedence.

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Solution 1 — A mass on a spring. Return →

(a) ω = √(k/m) = √(80/0.50) = 12.6 {rad,s^(-1)}, so T = 2π/ω = 0.50 {s} and f = 1/T = 2.0 {Hz}.

(b) v_(max) = ω A = (12.6)(0.12) = 1.5 {m,s^(-1)}, a_(max) = ω²A = (160)(0.12) = 19 {m,s^(-2)}.

(c) v = ω√(A²-x²) = 12.6√(0.12² - 0.06²) = 12.6(0.104) = 1.3 {m,s^(-1)}. **Check:** at x = A/2 the speed is (√(3))/(2)v_(max) = 0.866(1.5) = 1.3 {m,s^(-1)}, as expected from the sine of 60°.

Solution 2 — A pendulum clock. Return →

(a) T = 2π√(L/g) gives L = (gT²)/(4π²) = ((9.81)(4.000))/(39.48) = 0.994 {m}.

(b) T_({Moon}) = 2.000√((9.81)/(1.62)) = 2.000(2.46) = 4.92 {s} — the clock would run about two and a half times too slow.

(c) The clock loses 10 {s} in 86,400 {s}, a fractional gain of period 10/86,400 = 1.16×10^(-4). Since T∝√(L), the length must change by twice that fraction, 2.3×10^(-4) — about 0.23 {mm} on a metre, which is why the bob has a fine adjusting screw. **Check:** shortening the pendulum makes the period smaller, so the clock gains; the correction is to shorten it here.

Solution 3 — A physical pendulum. Return →

(a) $I = (1)/(3)mL^2 = (1)/(3)(0.60)(0.80)^2 = 0.128$ {kg,m²}, $d = 0.40$ {m}, so

$$T = 2\pi\sqrt{((I)/(mgd))} = 2\pi\sqrt{((0.128)/((0.60)(9.81)(0.40)))} = 2\pi\sqrt{(0.0544)} = 1.47 \text{ \{s\}}. \quad (1)$$

(b) A simple pendulum with the same period has length $= (I)/(md) = (0.128)/(0.24) = 0.533$ {m}, that is $(2)/(3)L$ as the text states.

(c) Pivoted at its centre, $d = 0$ and the gravitational torque vanishes for small displacements: there is no restoring force and no oscillation (the rod is in neutral equilibrium). The general formula shows this as $T \to \infty$. **Check:** $T = 1.47$ {s} is shorter than the 1.80 {s} of a simple pendulum of the full length 0.80 {m}, as a physical pendulum must be, since its mass is concentrated nearer the pivot than the bob of a "simple" pendulum.

Solution 4 — Energy in SHM. Return →

(a) $E = (1)/(2)kA^2 = (1)/(2)(200)(0.15)^2 = 2.25$ {J}.

(b) $U = (1)/(2)kx^2 = (1)/(2)(200)(0.05)^2 = 0.25$ {J}, so $K = 2.0$ {J}.

(c) $K = 3U$ together with $K+U = E$ gives $4U = E$, so $U = E/4$ and $x = A/2 = 7.5$ {cm} (on either side). **Check:** at $x = A/2$, $U = (1)/(4)E$ and $K = (3)/(4)E$; the ratio is indeed 3.

Solution 5 — Damping. Return →

(a) $\omega_0 = \sqrt{(k/m)} = \sqrt{(32/0.20)} = 12.6$ {rad,s^(-1)}; $\gamma = b/(2m) = 0.32/0.40 = 0.80$ {s^(-1)}; $Q = \omega_0/(2\gamma) = 12.6/1.6 = 7.9$.

(b) $\omega_d = \sqrt{(\omega_0^2-\gamma^2)} = \sqrt{(160 - 0.64)} = 12.6$ {rad,s^(-1)}, only 0.2% below ω_0: the frequency is very insensitive to weak damping, while the amplitude is not.

(c) $A = A_0e^{(-\gamma t)}$ gives $t = (\ln 10)/(\gamma) = (2.303)/(0.80) = 2.9$ {s}. Energy goes as the square of the amplitude, so the fraction left is $(0.1)^2 = 1\%$. **Check:** $Q = 7.9$ means about $Q/\pi = 2.5$ oscillations before the amplitude falls by e, and 2.9 {s} is indeed several periods of 0.5 {s}.

Solution 6 — A driven oscillator. Return →

(a) ω_0 = √(k/m) = √(40/0.10) = 20 {rad,s^(-1)}.

(b) At resonance,

$$A = (F_0/m)/(2\gamma\omega_0) = (5.0)/(2(1.0)(20)) = 0.125\ \{m\}. \tag{2}$$

At ω = 60 {rad,s^(-1)},

$$A = (5.0)/(\sqrt{((400-3600)^2 + 4(1)^2(3600))}) = (5.0)/(\sqrt{(1.024\times10^7)}) = 1.6\times10^{-3}\ \{m\}. \tag{3}$$

(c) When ω≫ω_0 the $(\omega_0^2-\omega^2)^2\approx\omega^4$ term dominates the denominator, so $A\approx F_0/(m\omega^2)$. The force reverses direction long before the mass can follow it, so the mass barely moves: the displacement is then just the acceleration F_0/m divided by ω². **Check:** the ratio of the two amplitudes is 0.125/0.0016 ≈ 80, close to Q-like enhancement plus the 1/ω² decay.

Solution 7 — A buoy in water. Return →

(a) Pushing the buoy down by x displaces an extra volume Ax, so the extra buoyancy is ρ gAx upwards:

$$M\ x = -\rho\ gAx\ \omega = \sqrt{((\rho\ gA)/(M))} = \sqrt{(((1000)(9.81)(0.20))/(60))} = \sqrt{(32.7)} = 5.7\ \{rad,s^{-1}\}, \tag{4}$$

so T = 1.1 {s}.

(b) For a large amplitude the sides of the cylinder may leave the water or the top may dip below the surface, so the restoring force stops being proportional to x; in the usual case where the buoy rises out of the water, the effective stiffness falls and the period **increases**.

(c) A wide buoy has a large metacentric height: the centre of buoyancy moves far when the buoy tilts, giving a large righting moment. A tall narrow one has its centre of mass high and its waterplane small, so a slight tilt moves the buoyancy centre very little and the righting moment is weak or even negative. **Check:** T = 1.1 {s} rises as 1/√(A) and falls as 1/√(M), both as intuition suggests.

Solution 8 — Two springs. Return →

(a) With one spring on each side, displacing the block by x stretches one by x and compresses the other by x; both push back, so the total restoring force is 2kx and the effective stiffness is 2k:

$$T = 2\pi\sqrt{(m)/(2k)}). \tag{5}$$

(b) Two springs in parallel, both attached to the same face, also give 2kx in total, so the period is the same, 2π√(m/(2k)). What matters is the **sum** of the restoring forces for a given displacement, not the geometry.

(c) A single spring of stiffness 2k is mechanically the same as the pair, so nothing would change. If instead one of the two springs is removed, the stiffness halves and the period grows by √(2). **Check:** with the two-spring arrangement the period is shorter than with one spring by exactly √(2), as the formula shows.

Solution 9 — The large-angle pendulum. Return →

(a) T_0 = 2π√(L/g) = 2π√(1.00/9.81) = 2.01 {s}.

(b) T≈ T_0(1 + (θ_0²)/(16)) with θ_0 = 60° = 1.047 {rad}:

$$T \approx 2.01(1 + (1.096)/(16)) = 2.01(1.0685) = 2.15\ \{s\}, \tag{6}$$

in agreement with the exact value: the correction accounts for the difference almost entirely.

(c) Energy: the bob is released a height h = L(1-cos θ_0) = 1.00(1-0.5) = 0.50 {m} above the bottom, so v = √(2gh) = √(9.81) = 3.13 {m,s^(-1)}. The SHM estimate v_(max) = ω A uses A = Lθ_0 = 1.047 {m} and ω = √(g/L) = 3.13 {rad,s^(-1)}, giving 3.28 {m,s^(-1)} — close, but too large, because SHM assumes the arc is a straight line while the real path is an arc of the **circle** and the real force is weaker than mgθ. **Check:** as θ_0→0 the two speeds converge, which is the signature of a correct small-angle approximation.

Solution 10 — Resonance in a real system. Return →

(a) Four springs in parallel: k_({tot}) = 4(2.5×10^4) = 1.0×10^5 {N,m^(-1)}, so

$$\omega_0 = \sqrt{(1.0\times10^5)/(1200)} = 9.1\ \{\mathrm{rad,s^{-1}}\},\ f_0 = 1.45\ \{\mathrm{Hz}\}. \tag{7}$$

(b) γ = b/(2m) = 1200/2400 = 0.50 {s^(-1)}, so Q = ω_0/(2γ) = 9.1 and = γ/ω_0 = 0.055: strongly underdamped, far from critical, which is why a car bounces a few times after a bump rather than settling instantly.

(c) The worst ride occurs when the ridge-passing frequency equals f_0: at speed v the frequency is v/4.0, so v = 4.0×1.45 = 5.8 {m,s^(-1)} ≈ 21 {km,h^(-1)}. Lowering Q (better shock absorbers) or changing k or m shifts or flattens the resonance; in practice suspension is designed so that the body frequency is around 1 {Hz} and well damped, while the wheels have a separate, higher resonance. **Check:** the number 1.45 {Hz} is typical of a real car, which supports the estimate.